
On the prediction of critical transitions



Candidate:
Davide Papapicco

Supervisors:
Graham Donovan, Lauren Smith

Department of Mathematics, Faculty of Science
Waipapa Taumata Rau, The University of Auckland

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Abstract

To be decided.

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To be decided.

Davide Papapicco
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Chapter 1

Introduction

Give an overview of what the thesis addresses and its structure.

1.1 Motivation

What are tipping points and what are EWS? Examples and applications in natural and human sciences in which these are present.

1.2 Context

Start with the early works in catastrophe theory (60s, 70s). Then jump to the early 2000s with the ecological and climate science (emphasize the renaissance moment of critical transitions as a renaissance of the theory). Then talk about the interest of the mathematical community in the 2010s.

1.3 The gap

Discuss the current state of the literature and how the thesis fits (interpretability and augmented usage for real world applications). Stress the disconnect between the mathematical and scientific community when it comes to early-warning signals.

Chapter 2

Background

Emphasize that not all critical transitions are predictable under the currently developed framework and cite criticism within the community. Stress how the current theory is largely developed for stochastically forced systems that experience a slow passage through a saddle-node bifurcation. The goal of this section is to select a candidate timeseries from the real world (given in Motivation) and fit a saddle-node model with stochastic fluctuations. To do so we need two main ingredients: geometric singular perturbation theory (or Fenichel theory) and large deviation theory (or Freidlin and Wentzell theory).

2.1 Nonautonomous dynamics

Introduce the autonomous saddle-node normal form and an example of coordinate transformations for the saddle-node bifurcation theorem. Extend it to the dynamic case with the notion of pullback attractors.

2.2 Stochastic calculus

Introduce the fundamentals of stochastic calculus including the OUP and the FPE.

2.3 Large deviations in slow-fast systems

State that the correct mathematical interpretation of critical transitions is that of rare events in small perturbations of dynamic saddle-node systems. Introduce the two ingredients separately.

2.3.1 Geometric singular perturbation theory

Main results of Fenichel theory.

2.3.2 Large deviation theory

Main results of Freidlin-Wentzell theory.

Chapter 3

Forecasting tipping times from data

Paper n.2 but without the applications.

Chapter 4

Likelihood estimation of an interpretable early-warning

Paper n.1 but without the applications.

Chapter 5

Applications

Discuss and present applications of paper 1 and paper 2. Apply both methodologies to the chosen timeseries from Motivation and Background.

Chapter 6

Conclusions

